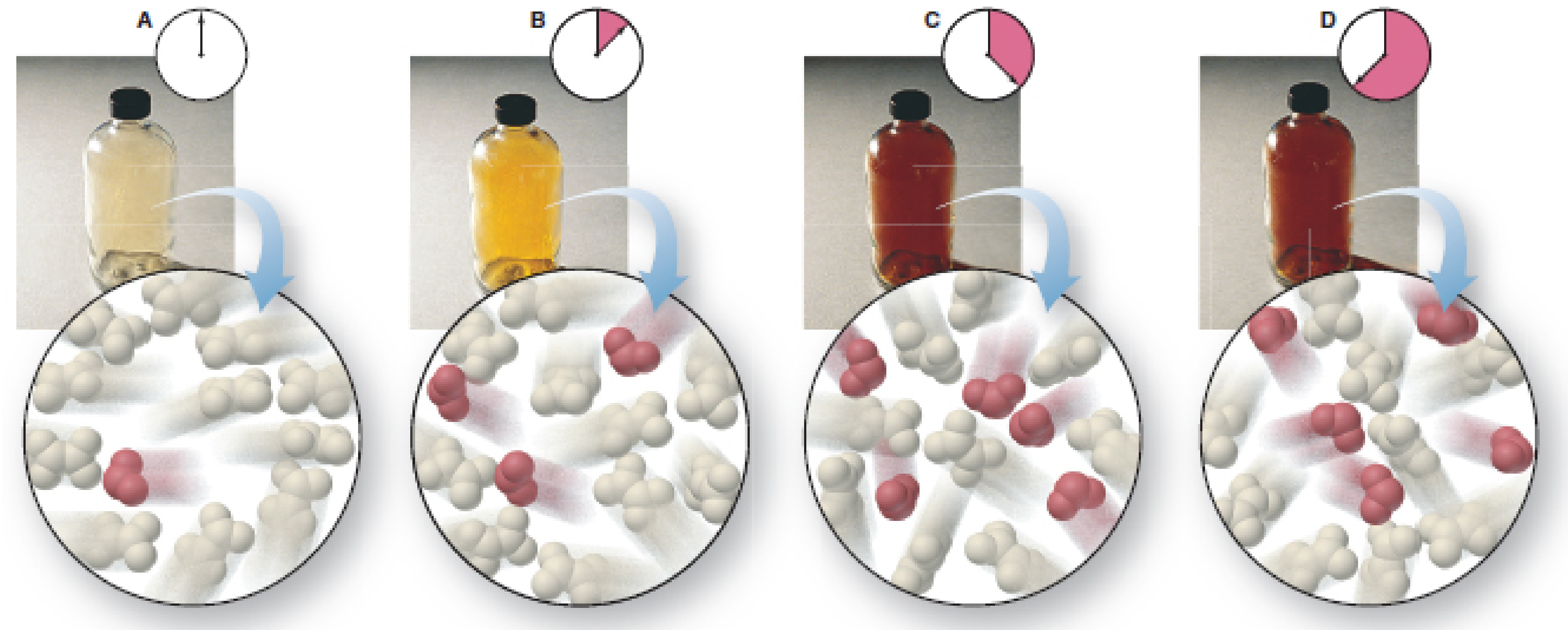
Chapter 17: Kinetics Content Lecture Pt. 3



KINETICS Part 3

Central focus:

The rates of chemical reactions

(How fast does the reaction proceed)

Chapter 17: Kinetics

17.1-17.2: Reaction Rates

17.3: Rate Law/Reaction Orders/Rate Constant

17.4:Integrated Rate Laws

17.5: Collision and Transition State Theory

17.6: Reaction Mechanisms

17.7: Catalysis

From reactant to products

The path of the reaction – sequence of single reaction steps that add up to give the overall equation – Reaction Mechanism…series of reaction steps

Overall Reaction

$$A + 2B \longrightarrow E + 2F$$

The above reaction may involve the following steps:

Elementary Reaction Steps

$$\sum \begin{array}{rcl} A + B & \longrightarrow & C \\ C + B & \longrightarrow & D + E \\ D & \longrightarrow & 2F \\ \hline A + 2B & \longrightarrow & E + 2F \end{array}$$

Overall Reaction

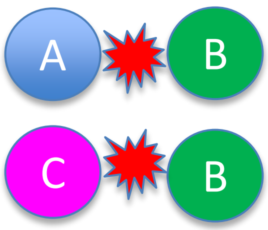
We propose mechanism(s) – but it (they) must be confirmed with the observed rate law/experimental observations

Elementary Reactions

The path of the reaction – sequence of single reaction steps that add up to give the overall equation – Reaction Mechanism

$$A + 2B \longrightarrow E + 2F$$

The above reaction may involve the following steps:



$$A + B \longrightarrow C$$

$$C + B \longrightarrow D + E$$

$$D \longrightarrow 2F$$

Individual steps of a reaction mechanism that describe a single molecular event – ELEMENTARY REACTIONS

Elementary reactions reflect the actual molecules coming together

Elementary reactions are usually unimolecular or bimolecular

Why?

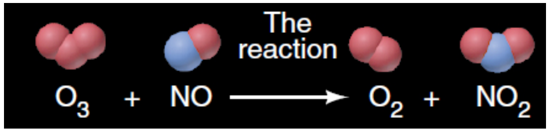
Elementary Reactions

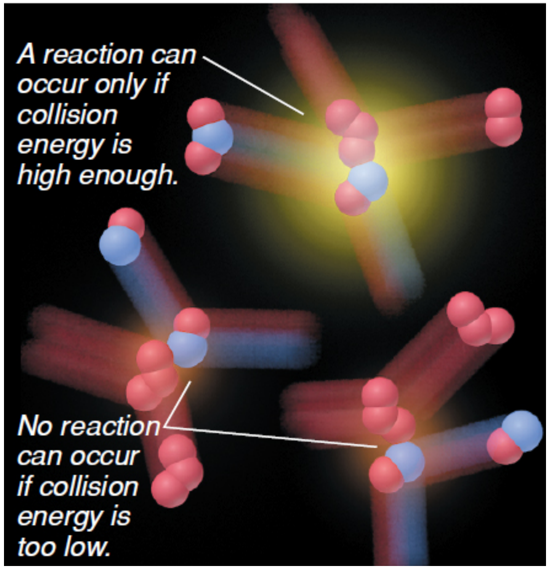
Elementary reactions reflect the actual molecules coming together

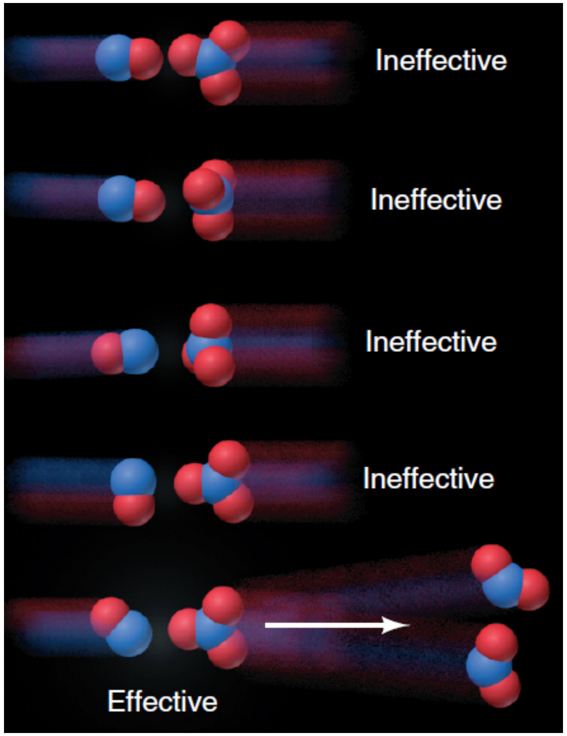
Elementary reactions are usually unimolecular or bimolecular

Why?

It is **highly unlikely** that 3 or more molecules would collide at the same time, with enough energy, and with the proper orientations for reaction







Elementary Reactions

Elementary reactions can be categorized based on the reactant stoichiometry

$$A + 2B \longrightarrow E + 2F$$

If below is the reaction mechanism (confirmed by rate law)

Molecularity determined by the reactants – uni – one; bi - two

$$\begin{array}{lllll} & A + B & \longrightarrow & \cancel{C} & \text{Bimolecular} \\ & \cancel{C} + B & \longrightarrow & \cancel{D} + E & \text{Bimolecular} \\ \Sigma & \cancel{D} & \longrightarrow & 2F & \text{Unimolecular} \\ \hline & A + 2B & \longrightarrow & E + 2F & \end{array}$$

1) The sum of the individual elementary reactions must add up to give the overall reaction

Elementary Reactions

Elementary reactions can be categorized based on the reactant stoichiometry

$$A + 2B \longrightarrow E + 2F$$

If below is the reaction mechanism (confirmed by rate law)

Mechanism	Molecularity	Rate Law
$A + B \xrightarrow{k_1} C$	Bimolecular	k_1 [A] [B]
$C + B \xrightarrow{k_2} D + E$	Bimolecular	k_2 [C] [B]
$D \xrightarrow{k_3} 2F$	Unimolecular	k_3 [D]

2) Rate law for each step can be deduced directly from reaction stoichiometry **but only for elementary reactions**

Elementary Reactions

Elementary reactions can be categorized based on the reactant stoichiometry

$$A + 2B \longrightarrow E + 2F$$

If below is the reaction mechanism (confirmed by rate law)

	Molecularity
$A + B \longrightarrow C$	Bimolecular
$C + B \longrightarrow D + E$	Bimolecular
$D \longrightarrow 2F$	Unimolecular

3) Elementary reaction is reversible

Elementary Reactions

Elementary reactions can be categorized based on the reactant stoichiometry

$$A + 2B \longrightarrow E + 2F$$

If below is the reaction mechanism (confirmed by rate law)

	Molecularity
$A + B \longrightarrow C$	Bimolecular
$C + B \longrightarrow D + E$	Bimolecular
$D \longrightarrow 2F$	Unimolecular

C & D are intermediates

4) Intermediates – (if applicable) produced in one elementary step and consumed in a subsequent step – are not present in the overall reaction equation

Elementary Reactions

Elementary reactions can be categorized based on the reactant stoichiometry

$$A + 2B \longrightarrow E + 2F$$

If below is the reaction mechanism (confirmed by rate law)

			Molecularity	Rate Law	
A + B	$\xrightarrow{k_1}$	C	Bimolecular	$k_1[A]\cdot[B]$	Slow
C + B	$\xrightarrow{k_2}$	D + E	Bimolecular	$k_2[C]\cdot[B]$	Fast
D	$\xrightarrow{k_3}$	2F	Unimolecular	$k_3[D]$	Fast

5) If one step is much slower than the rest of the steps – rate determining step.

You only include elementary reactions up to the slow rate determining step
In determining the overall rate law for a reaction

$$\boxed{R = k_1[A]\cdot[B]}$$

Initial elementary step is rate-determining

$$H_{2(g)} + 2ICl(g) \longrightarrow I_{2(g)} + 2\,HCl(g)$$

Proposed reaction mechanism:

SLOW ← rate determining

Rate

Step 1: $H_{2(g)} + ICl(g) \xrightarrow{k_1} \cancel{HI}(g) + HCl(g)$ $\qquad k_1[H_2]\cdot[ICl]$

FAST

Step 2: $\cancel{HI}(g) + ICl(g) \rightleftharpoons I_{2(g)} + HCl(g)$

$$Rate = k_1[H_2]\cdot[ICl]$$

This rate law is confirmed by experiments

1. Check if the elementary steps add to give the overall reaction ✓
2. Determine the rate law based on each elementary step ✓
3. Determine if this calculated rate law matches with experiments ✓

Initial elementary step is rate-determining

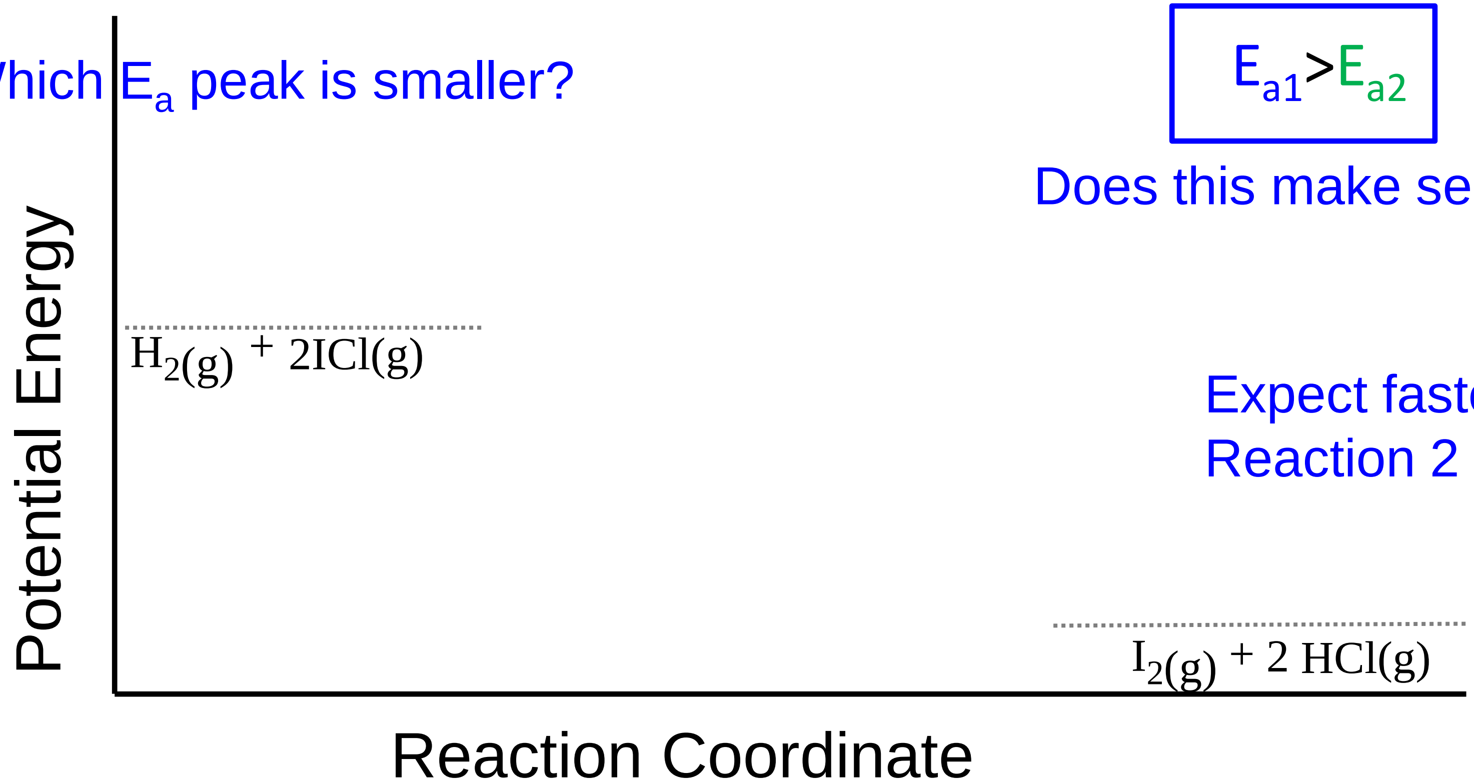
$$H_{2(g)} + 2ICl(g) \longrightarrow I_{2(g)} + 2\,HCl(g)$$

Proposed reaction mechanism:

Step 1: $H_{2(g)} + ICl(g) \xrightarrow[]{\text{SLOW}\ k_1} HI(g) + HCl(g)$

Step 2: $HI(g) + ICl(g) \xrightleftharpoons{\text{FAST}} I_{2(g)} + HCl(g)$

Which E_a peak is smaller?



$\boxed{E_{a1} > E_{a2}}$

Does this make sense?

Expect faster Reaction 2

Fast Initial step

$$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$$

Proposed reaction mechanism:

Rate

Fwd Rev

Step 1: $NO(g) + O_2(g) \underset{}{\overset{\text{FAST}}{\rightleftharpoons}} \cancel{NO_3}(g)$

NO_3 is an intermediate

Step 2: $\cancel{NO_3}(g) + NO(g) \xrightarrow{\text{SLOW}} 2NO_2(g)$

1. Check if the elementary steps add to give the overall reaction ✓
2. Determine the rate law based on each elementary step Include all steps up to slow rxn
3. Determine if this calculated rate law matches with experiments

Fast Initial step

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2\,(g)$$

Rate Law $= k[NO]^2[O_2]$

Proposed reaction mechanism:

	Rate Fwd	Rate Rev
Step 1 (FAST): $NO(g) + O_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NO_3\,(g)$	$k_1\,[NO]\,[O_2]$	$k_{-1}\,[NO_3]$

Step 1 reaches quasi equilibrium

Forward rate = Reverse rate

$$k_1\,[NO]\,[O_2] = k_{-1}\,[NO_3]$$

Isolate the intermediate → $[NO_3] = (k_1/k_{-1})[NO]\,[O_2]$

Substitute in

Step 2 (SLOW): $NO_3\,(g) + NO\,(g) \xrightarrow{k_2} 2\,NO_2\,(g)$

Rate determining step

$$k_2[NO_3]\,[NO]$$

$$k_2(k_1/k_{-1}[NO]\,[O_2])[NO]$$

$$(k_2k_1/k_{-1})[NO]^2[O_2]$$

Remember that Rate Laws cannot contain Intermediates in the final equation

$$\boxed{Rate = k[NO]^2 \cdot [O_2] \quad \text{where } k = \frac{k_2k_1}{k_{-1}}}$$

Fast Initial-Step

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2\,(g)$$

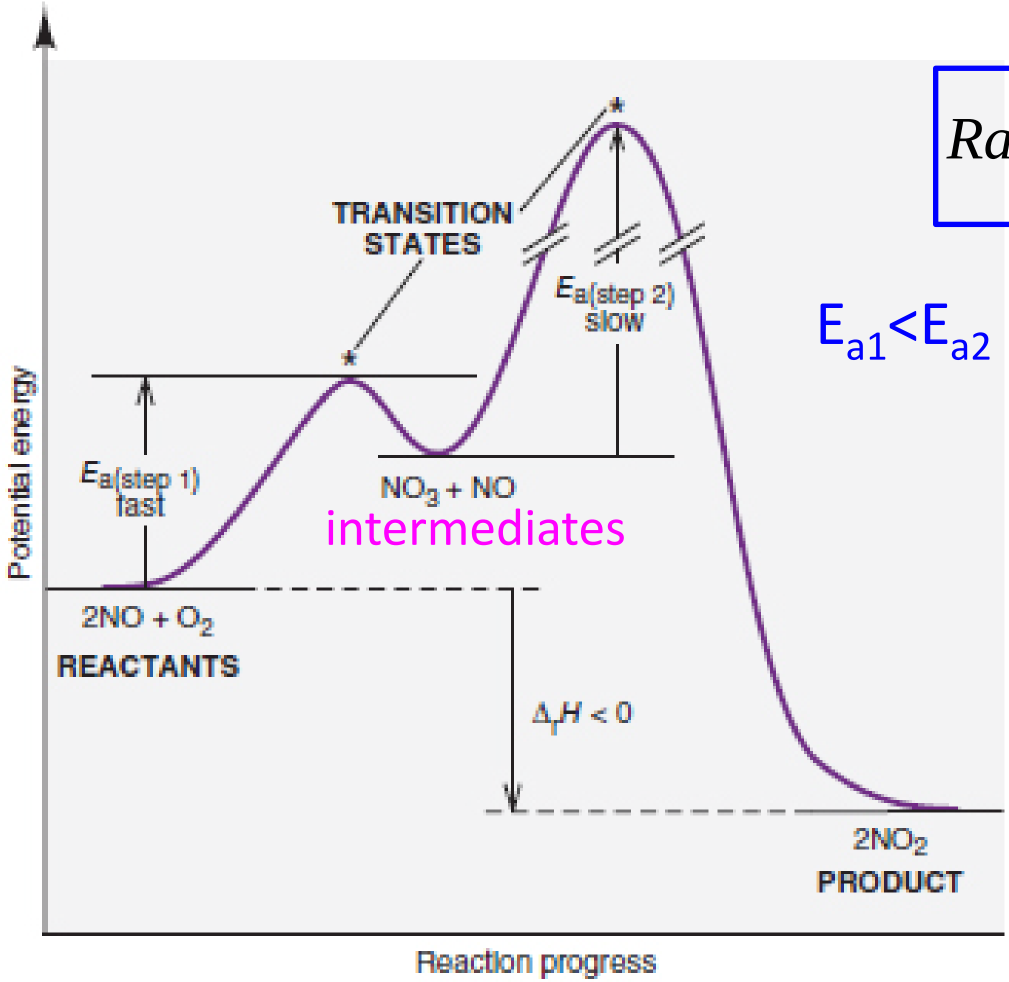
FAST

$$NO(g) + O_2(g) \rightleftharpoons NO_3\,(g)$$

SLOW

$$(g) \longrightarrow 2\,NO_2\,(g)$$

$$Rate = k[NO]^2 \cdot [O_2] \quad \text{where } k = \frac{k_2 k_1}{k_{-1}}$$



B

STEADY STATE APPROXIMATIONS

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$$

Proposed reaction mechanism:

Step 1: $NO(g) + O_2(g) \xrightarrow{k_1} NO_3(g)$

Step 2: $NO_3(g) + NO(g) \xrightarrow{k_2} 2\,NO_2(g)$

Sometimes we do not know which of the above steps is the rate determining (slow)… – how do you determine it?

Use the steady state approximation

STEADY STATE APPROXIMATIONS

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2\,(g)$$ Observed rate: $k\,[NO]^2[O_2]$

Proposed reaction mechanism:

Step 1: $NO(g) + O_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NO_3\,(g)$

Step 2: $NO_3\,(g) + NO\,(g) \xrightarrow{k_2} 2\,NO_2\,(g)$

Assume the first reaction is
Reversible in order to
Achieve a steady state in
The intermediate

Use the steady state approximation – The concentration of the intermediate species remains constant throughout the reaction

Rate at which the intermediate is produced = rate at which the intermediate is consumed

Use the steady state approximation and the experimentally determined rate law to determine which elementary step is the rate determining step

STEADY STATE APPROXIMATIONS

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$$

Proposed reaction mechanism:

Step 1: $NO(g) + O_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NO_3(g)$

Step 2: $NO_3(g) + NO(g) \xrightarrow{k_2} 2\,NO_2(g)$

$$\frac{d[NO_3]}{dt} = 0$$

This is the steady state approximation

(rate of formation of NO_3 = rate of consumption of NO_3)

Rate of formation of NO_3 =

Which reactions are forming NO_3?

Rate of consumption of NO_3 =

Which reactions are consuming NO_3?

STEADY STATE APPROXIMATIONS

$$2\ NO(g) + O_2(g) \longrightarrow 2\ NO_2\ (g)$$

Proposed reaction mechanism:

Step 1: $NO(g) + O_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NO_3\ (g)$

Step 2: $NO_3\ (g) + NO\ (g) \xrightarrow{k_2} 2\ NO_2\ (g)$

$$\frac{d[NO_3]}{dt} = 0$$

This is the steady state approximation

(rate of formation of NO_3 = rate of consumption of NO_3)

Rate of formation of NO_3 =

$$k_1[NO]\cdot[O_2]$$

Rate of consumption of NO_3 =

$$k_{-1}[NO_3]+k_2[NO_3][NO]$$

$$k_1[NO]\cdot[O_2]=k_{-1}[NO_3]+k_2[NO_3][NO]$$

Solve for $[NO_3]$

STEADY STATE APPROXIMATIONS

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$$

Proposed reaction mechanism:

Step 1: $NO(g) + O_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NO_3(g)$

Step 2: $NO_3(g) + NO(g) \xrightarrow{k_2} 2\,NO_2(g)$

$$[NO_3] = \frac{k_1[NO]\cdot[O_2]}{(k_{-1} + k_2[NO])}$$

STEADY STATE APPROXIMATIONS

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$$

Proposed reaction mechanism:

Rate

Step 1: $NO(g) + O_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NO_3(g)$ $k_1[NO][O_2]$

Step 2: $NO_3(g) + NO(g) \xrightarrow{k_2} 2\,NO_2(g)$ $k_2[NO_3][NO]$

Write the elementary rates for each step and substitute for intermediate

$$[NO_3] = \frac{k_1[NO]\cdot[O_2]}{(k_{-1} + k_2[NO])}$$

Substitute for $[NO_3]$

Rate (Step 1) $k_1[NO][O_2]$

Rate (Step 2) $k_2[NO_3][NO] = k_2 \dfrac{k_1[NO][O_2]}{k_{-1} + k_2[NO]}[NO]$

STEADY STATE APPROXIMATIONS

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$$

Observed rate: $k\,[NO]^2[O_2]$

Proposed reaction mechanism:

		Rate
Step 1:	$NO(g) + O_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NO_3(g)$	$k_1[NO][O_2]$
Step 2:	$NO_3(g) + NO(g) \xrightarrow{k_2} 2\,NO_2(g)$	$k_2[NO_3][NO]$

$$[NO_3] = \frac{k_1[NO]\cdot[O_2]}{(k_{-1} + k_2[NO])}$$

Compare this to experimentally determined rate law: $k\,[NO]^2[O_2]$

Rate (Step 1) $\quad k_1[NO][O_2]$

Rate (Step 2) $\quad k_2[NO_3][NO] = k_2 \dfrac{k_1[NO][O_2]}{k_{-1} + k_2[NO]}[NO]$

STEADY STATE APPROXIMATIONS

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$$

Observed rate: $k\,[NO]^2[O_2]$

Proposed reaction mechanism:

		Rate
Step 1:	$NO(g) + O_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NO_3(g)$	$k_1[NO][O_2]$
Step 2:	$NO_3(g) + NO(g) \xrightarrow{k_2} 2\,NO_2(g)$	$k_2[NO_3][NO]$

$$[NO_3] = \frac{k_1[NO]\cdot[O_2]}{(k_{-1} + k_2[NO])}$$

Rate (Step 2)

$$k_2[NO_3][NO] = k_2\,\frac{k_1\,[NO]\,[O_2]}{k_{-1} + k_2[NO]}\,[NO]$$

We can now consider two assumptions – which will enable us to compare Step 1 and Step 2:
Is the rate of NO_3 consumption faster in step 1 or step 2:

Is $k_{-1}[NO_3] > k_2\,[NO]\,[NO_3]$ **or** $k_{-1}[NO_3] < k_2\,[NO]\,[NO_3]$
Which simplifies to: Is $k_{-1} > k_2\,[NO]$ or $k_{-1} < k_2\,[NO]$

STEADY STATE APPROXIMATIONS

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2\,(g)$$

Observed rate: $k\,[NO]^2[O_2]$

Proposed reaction mechanism:

		Rate
Step 1:	$NO(g) + O_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NO_3\,(g)$	$k_1[NO][O_2]$
Step 2:	$NO_3\,(g) + NO\,(g) \xrightarrow{k_2} 2\,NO_2\,(g)$	$k_2[NO_3][NO]$

If $k_{-1} <<< k_2[NO]$

$$[NO_3] = \frac{k_1[NO]\cdot[O_2]}{\left(k_{-1} + k_2[NO]\right)} \approx \frac{k_1[NO]\cdot[O_2]}{\left(k_2[NO]\right)}$$

Rate (Step 2)

$$k_2[NO_3][NO] = k_2\,\frac{k_1\,[NO]\,[O_2]}{k_{-1} + k_2[NO]}\,[NO]$$

$$\cancel{k_2}\,\frac{k_1\,[\cancel{NO}]\,[O_2]}{\cancel{k_2}[\cancel{NO}]}\,[NO]$$

$$\downarrow$$

$$k_1\,[NO]\,[O_2]$$

This rate is the same as if step 1 was the slow step.
Does not match the exp. rate law
So this **is not the rate determining step**

STEADY STATE APPROXIMATIONS

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$$

Observed rate: $k\,[NO]^2[O_2]$

Proposed reaction mechanism:

		Rate
Step 1:	$NO(g) + O_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NO_3(g)$	$k_1[NO][O_2]$
Step 2:	$NO_3(g) + NO(g) \xrightarrow{k_2} 2\,NO_2(g)$	$k_2[NO_3][NO]$

If $k_{-1} >>> k_2[NO]$

$$[NO_3] = \frac{k_1[NO]\cdot[O_2]}{(k_{-1} + k_2[NO])} \approx \frac{k_1[NO]\cdot[O_2]}{k_{-1}}$$

Rate (Step 2)

$$k_2[NO_3][NO] = k_2\,\frac{k_1\,[NO]\,[O_2]}{k_{-1} + k_2[NO]}\,[NO] \longrightarrow k_2\,\frac{k_1\,[NO]\,[O_2]}{k_{-1}}\,[NO]$$

$$Rate = k[NO]^2\cdot[O_2] \quad \text{where } k = \frac{k_2 k_1}{k_{-1}}$$

Matches with the exp. rate law so step 2 must be the slow step

STEADY STATE APPROXIMATIONS

$$H_2\,(g) + Br_2\,(g) \longrightarrow 2HBr$$ Overall Reaction

$$R = \frac{k[H_2][Br_2]^{\frac{1}{2}}}{1+\frac{k'[HBr]}{[Br_2]}}$$ Measured Rate Law

k

k' Two Rate Constants

Proposed Mechanism has 5 elementary reactions

Two steady state assumptions for intermediates!

M. Polanyi (father of Nobel prize winner M. Polanyi at U ofT)
Independently solved it

I would not ask a mechanism this complex on midterm or exam

STEADY STATE APPROXIMATIONS

- Connection between Kinetics & Equilibrium

$$A + B \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} P$$

Reaches Equilibrium

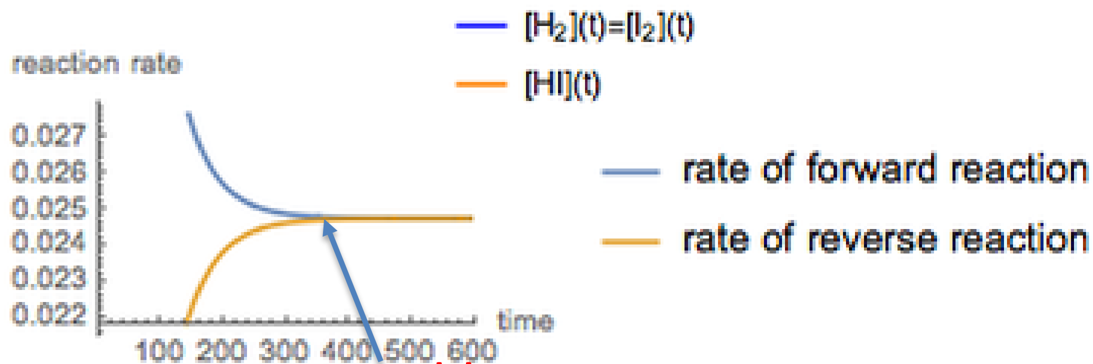
At Equilibrium

The rates of the forward and reverse reactions are equal

$$k_1[A][B] = k_{-1}[P]$$

$$K_{eq} = \frac{k_1}{k_{-1}} = \frac{[P]_{eq}}{[A]_{eq}[B]_{eq}}$$

$$H_2(g) + I_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2HI\,(g)$$



At Equilibrium

The concentrations stay constant

The rates are balanced

Does the reaction Stop at equilibrium?

Chapter 17: Kinetics

17.1-17.2: Reaction Rates

17.3: Rate Law/Reaction Orders/Rate Constant

17.4:Integrated Rate Laws

17.5: Collision and Transition State Theory

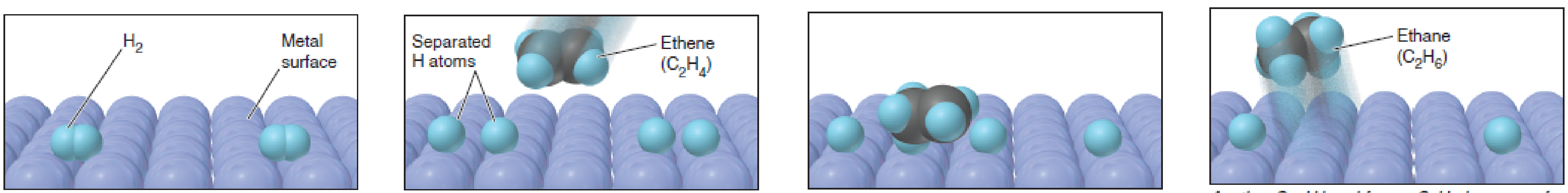
17.6: Reaction Mechanisms

17.7: Catalysis

Catalysis

Catalyst:

- Accelerates a reaction
- Without being consumed in the reaction (is not a part of the rate law)



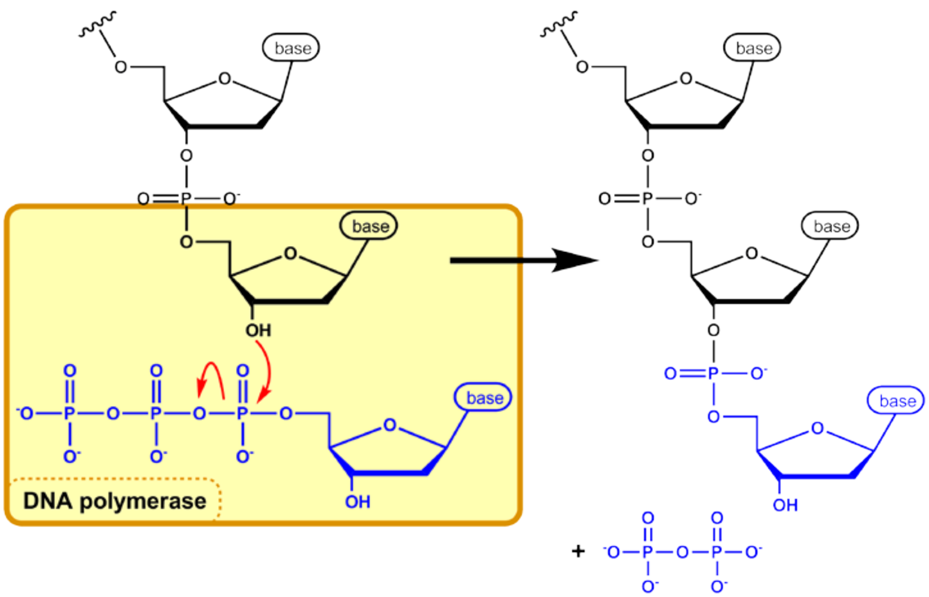
H_2 adsorbs to metal surface. *Rate-limiting step is H—H bond breakage.* *After C_2H_4 adsorbs, one C—H forms.* *Another C—H bond forms; C_2H_6 leaves surface*

Metal (usually Ni/Pt metal) acts as a catalyst for the hydrogenation of alkenes

Catalysis

Catalyst:

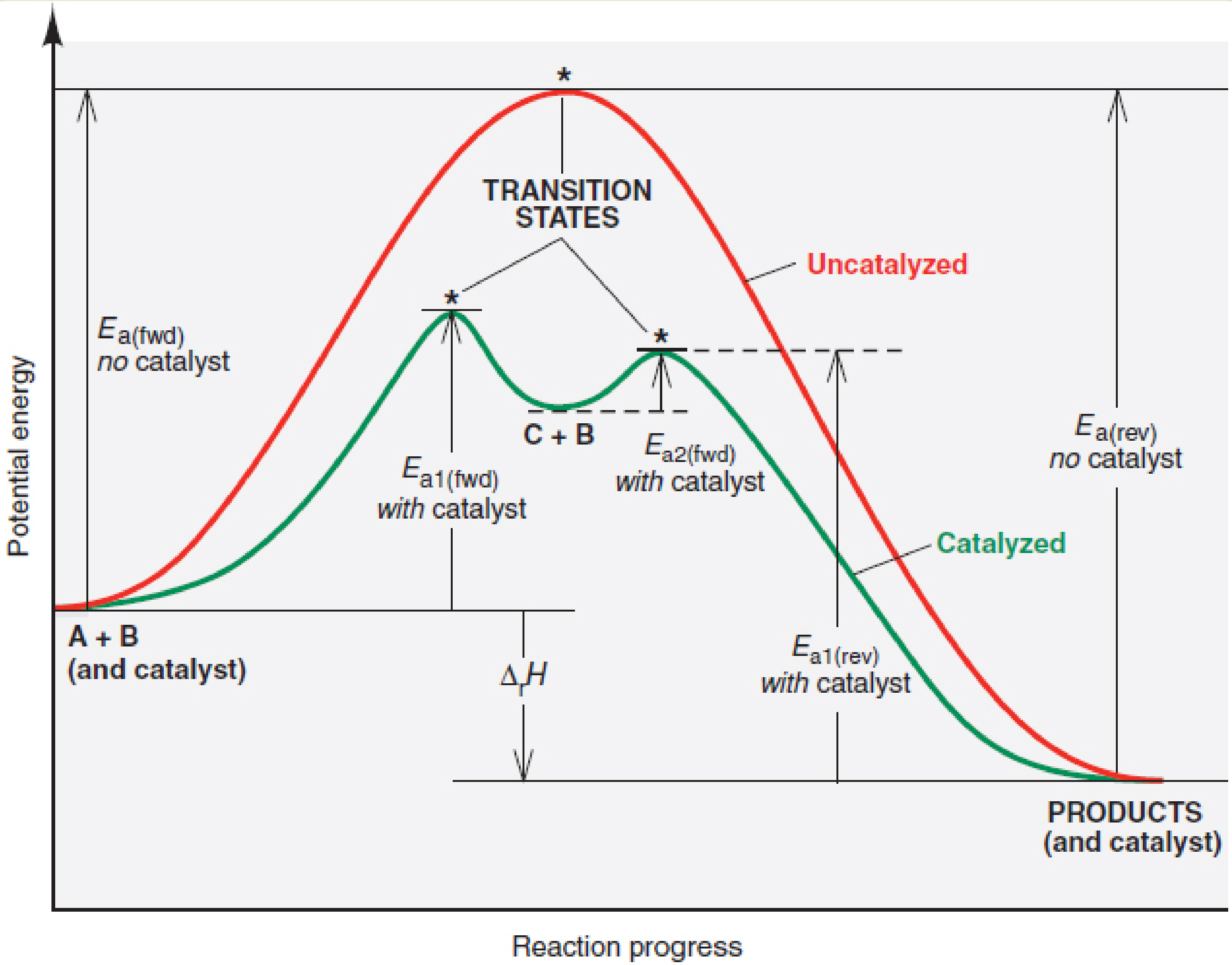
- Accelerates a reaction
- Without being consumed in the reaction (is not a part of the rate law)



DNA polymerase is a biological catalyst (enzyme)

Enzymes are proteins that act as catalysts in cells

Catalytic Action



A catalyst accelerates the reaction by providing an alternative reaction pathway with a **lower activation energy**

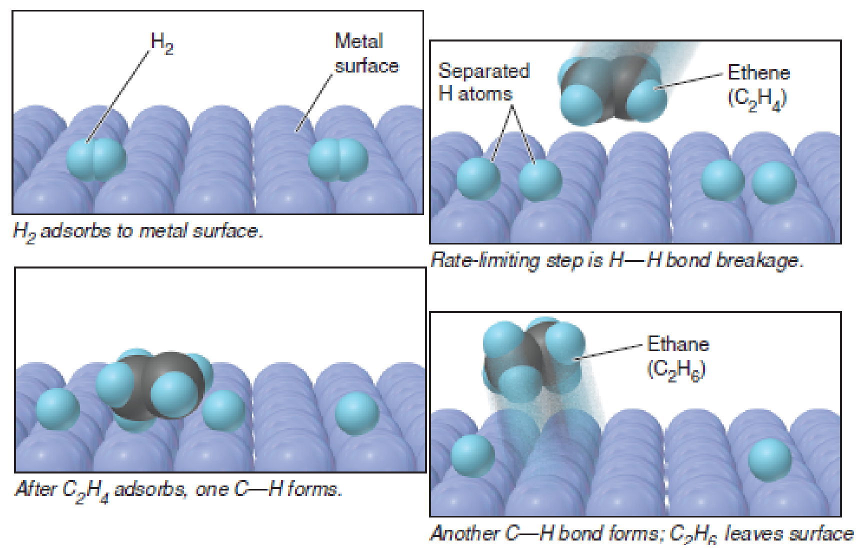
Homogeneous and Heterogeneous Catalysis

Homogeneous Catalysis
Catalysts is in the same phase as the reactants

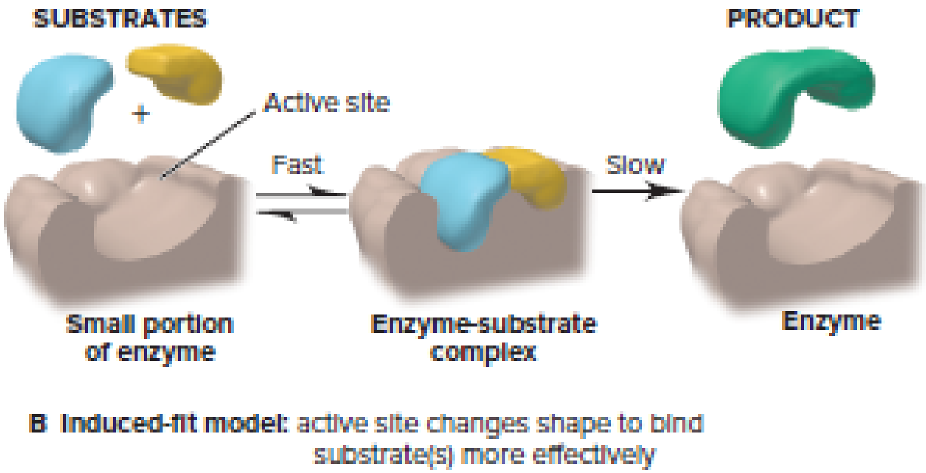
A

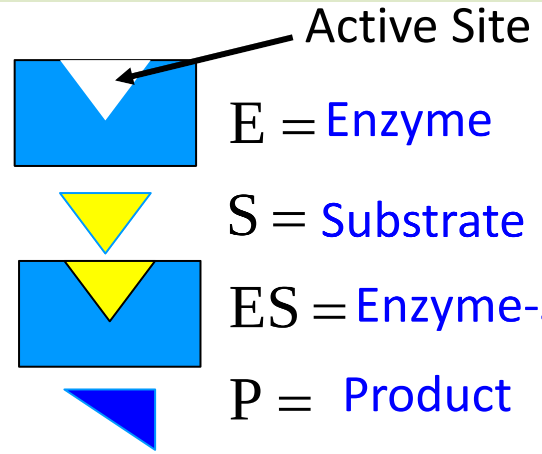
B

Heterogeneous Catalysis
Catalysts is in a different phase (usually solid)
Adsorption on the surface of the catalyst
Catalytic surface provides an active site for the reaction



Biological Catalysts: Enzymes



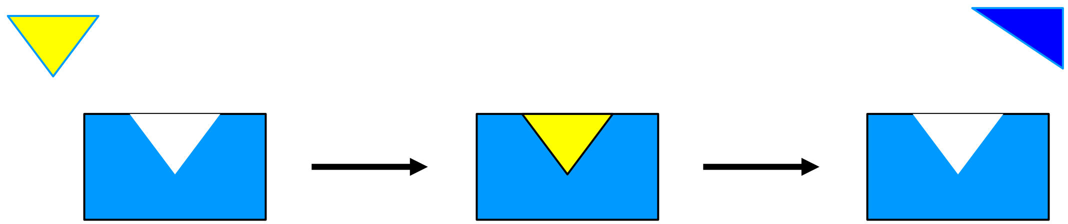


E = Enzyme

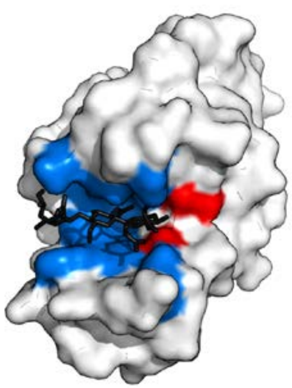
S = Substrate

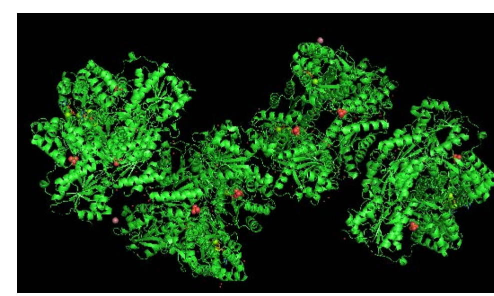
ES = Enzyme-Substrate complex

P = Product

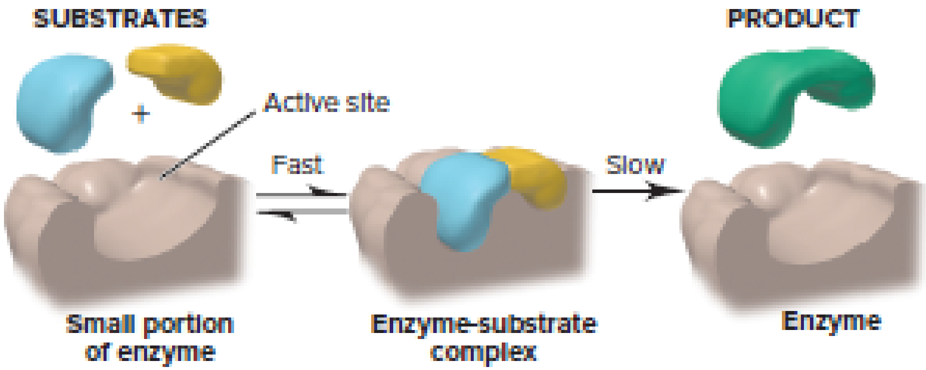


One enzymatic catalytic cycle…one turnover



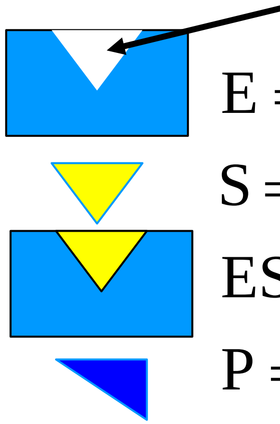


Biological Catalysts: Enzymes



B Induced-fit model: active site changes shape to bind substrate(s) more effectively

Active Site



E = Enzyme

S = Substrate

ES = Enzyme-Substrate complex

P = Product

Use steady-state approximation for ES

$$\mathrm{E} + \mathrm{S} \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} \mathrm{ES}$$

$$\mathrm{ES} \xrightarrow{k_2} \mathrm{E} + \mathrm{P}$$

$$\frac{d[\mathrm{ES}]}{dt} = 0$$

$[\mathrm{ES}] \approx$ Constant

Michaelis-Menten Kinetics

Excellent Wolfram Demonstration for Enzyme Kinetics

http://demonstrations.wolfram.com/MichaelisMentenEnzymeKineticsAndTheSteadyStateApproximation/

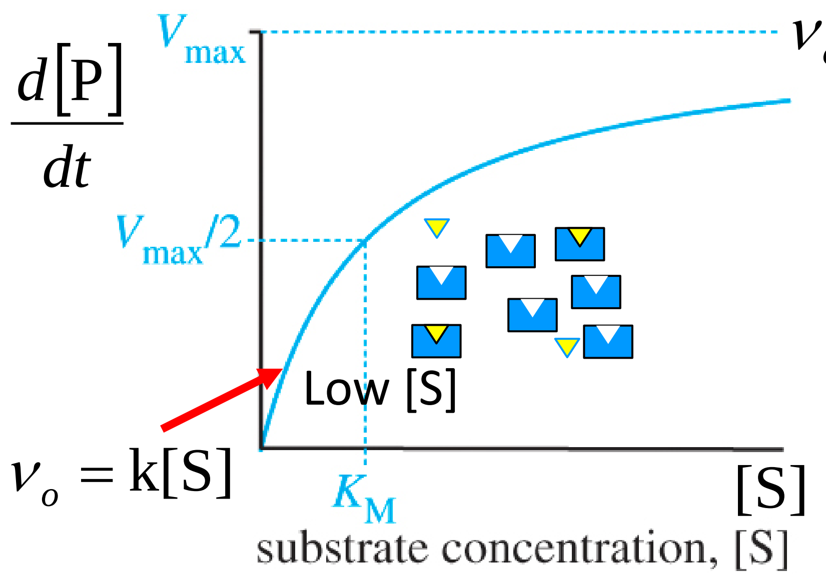
Michaelis-Menten Kinetics in Enzymes

$$\boxed{\frac{d[P]}{dt} = \frac{k_2[E]_0[S]}{K_M + [S]} \quad \text{where } K_M = \frac{k_{-1} + k_2}{k_1}}$$

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES$$

$$ES \xrightarrow{k_2} E + P$$

K_M is Michaelis Menten constant



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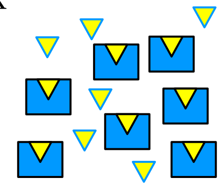
High [S]



All enzyme is occupied
Substrate is in excess
Need to wait for free
Active site
Maximum reaction rate